

Attention-Deficit/Hyperactivity Disorder and Late-Life Cognitive Impairment

Guilherme Bauer-Negrini, PhD; Douglas Teixeira Leffa, MD, PhD; Pamela C. L. Ferreira, PhD; Bruna Bellaver, PhD; Guilherme Povala, PhD; Marina S. Medeiros, MD; Firoza Z. Lussier, MS; Dana L. Tudorascu, PhD; Samuele Cortese, MD, PhD; Luis Augusto Rohde, MD, PhD; Brooke S. G. Molina, PhD; Tharick A. Pascoal, MD, PhD

[+ Supplemental content](#)

IMPORTANCE Attention-deficit/hyperactivity disorder (ADHD) affects tens to hundreds of millions of adults worldwide, and preliminary evidence suggests it may be associated with increased risk of mild cognitive impairment (MCI) and dementia. However, concerns regarding diagnostic ascertainment and residual confounding remain.

OBJECTIVE To investigate the association of ADHD with incident cognitive impairment, defined as MCI or dementia, using electronic health record (EHR) diagnoses and genetic liability measured using genome-wide ADHD polygenic risk score (ADHD PRS) in the same individuals.

DESIGN, SETTING, AND PARTICIPANTS This cohort study used data from the US All of Us Research Program (enrollment May 6, 2018, to October 1, 2023; EHR data from January 4, 1985, to October 1, 2023; mean [SD] follow-up 10.26 [6.75] years). A total of 187 341 participants aged 50 years and older with available EHR and genomic data were included. Data were analyzed from October 2025 to July 2026.

EXPOSURES ADHD diagnosis and ADHD PRS.

MAIN OUTCOMES AND MEASURES Incident MCI or dementia.

RESULTS Among 187 341 participants (mean [SD] age, 56.41 [7.88] years; 107 266 [57%] female), 3026 (1.6%) had ADHD. Among participants with ADHD, 203 (6.7%) developed cognitive impairment, compared with 6259 (3.4%) among those without ADHD. In adjusted Cox models, ADHD was associated with a more than 4-fold higher hazard of cognitive impairment (hazard ratio [HR], 4.60; 95% CI, 3.99-5.29; $P < .001$). Adjustments for comorbidities, educational attainment, and health care utilization attenuated but did not eliminate the association. Similarly, ADHD PRS was associated with incident cognitive impairment (highest vs lowest quintile: HR, 1.35; 95% CI, 1.25-1.46; $P < .001$). Among participants who developed cognitive impairment, comorbid ADHD diagnosis was associated with a nearly 2-fold higher odds of suboptimal mental health (odds ratio [OR], 1.69; 95% CI, 1.18-2.39; $P = .01$), quality of life (OR, 1.66; 95% CI, 1.14-2.39; $P = .02$), social satisfaction (OR, 2.39; 95% CI, 1.69-3.37; $P < .001$), and 20% higher health care utilization (incidence rate ratio, 1.20; 95% CI, 1.09-1.34; $P = .003$), suggesting greater care burden beyond the elevated incidence risk.

CONCLUSIONS AND RELEVANCE In this cohort study, convergent clinical and genetic evidence revealed ADHD was associated with a markedly increased risk of late-life cognitive impairment. Among those who develop cognitive impairment, an ADHD diagnosis was associated with worse health outcomes and greater health care utilization, supporting the need for targeted monitoring and integrated care in this population.

JAMA Psychiatry. doi:10.1001/jamapsychiatry.2026.3165
Published online October 7, 2026.

Author Affiliations: Author affiliations are listed at the end of this article.

Corresponding Author: Tharick A. Pascoal, MD, PhD, Department of Psychiatry, University of Pittsburgh, 200 Meyran Ave, Pittsburgh, PA 15213 (pascoal@upmc.edu).

Attention-deficit/hyperactivity disorder (ADHD) is a neurodevelopmental disorder characterized by pervasive and impairing symptoms of inattention, hyperactivity-impulsivity, or both.¹ Although historically viewed as restricted to childhood, ADHD is now recognized as a lifelong disorder that can persist across the lifespan, with population estimates suggesting a prevalence of approximately 1% among adults older than 60 years.² As populations age globally, the absolute number of older adults living with ADHD is expected to increase.³ However, ADHD remains relatively underinvestigated in later life compared with younger age groups.⁴

Of particular interest is the potential association between ADHD and late-life incidence of cognitive impairment. Emerging epidemiologic evidence suggests that ADHD may be associated with an increased risk of mild cognitive impairment (MCI) and dementia.⁵⁻¹² Alongside studies based on clinical diagnoses, recent research suggests an association between the ADHD polygenic risk score (ADHD PRS),¹³ a probabilistic estimate of ADHD risk based on population-level DNA association,¹⁴ and cognitive decline in older age.¹⁵ If ADHD contributes to risk of cognitive impairment, clarifying this relationship could have implications for risk stratification and prevention efforts.

Despite accumulating evidence, important limitations constrain the interpretation and generalizability of studies linking ADHD to MCI and dementia. Most reports rely solely on electronic health records (EHRs) to ascertain ADHD diagnosis, which may reflect differential health care access and use, clinician recognition, and documentation practices, potentially introducing bias.¹⁶ Additionally, adjustment for confounders has varied across studies.⁵⁻¹² Beyond risk of MCI and dementia, ADHD has been associated with worse perceived health,¹⁷ greater functional burden,¹⁸ and increased health care utilization.¹⁹ However, whether ADHD is associated with these outcomes among older adults with MCI or dementia remains unclear. These gaps motivate further investigation in diverse cohorts and approaches that expand EHR diagnoses with additional information, such as the ADHD PRS. To our knowledge, no prior study has evaluated whether EHR-documented ADHD and ADHD PRS provide complementary information about risk for MCI and dementia or examined self-reported health status and health care utilization among these individuals.

Therefore, this study had the following aims: (1) to investigate the association between ADHD diagnosis and incident MCI or dementia in a large, diverse sample from the All of Us Research Program while evaluating potential confounders; (2) to complement EHR-based ADHD ascertainment by analyzing genetic liability to ADHD using an ADHD PRS; and (3) to examine whether ADHD is associated with worse self-reported health status and health care utilization among individuals with cognitive impairment.

Methods

Study Design and Population

This retrospective cohort study used data from the All of Us Research Program^{20,21} (controlled tier version 8 [C2024Q3R8],

Key Points

Question Are diagnosis and polygenic risk of attention-deficit/hyperactivity disorder (ADHD) associated with increased risk of late-life cognitive impairment?

Findings In this cohort study of 187 341 adults from the All of Us Research Program with electronic health records and genomic data, ADHD was also associated with a higher hazard of incident mild cognitive impairment or dementia, and a higher ADHD polygenic risk score was associated with an increased risk.

Meaning ADHD, whether identified clinically or genetically, is associated with greater risk for late-life cognitive impairment.

cutoff date October 1, 2023), and followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.²² All participants provided written informed consent, and protocols were reviewed by the institutional review board of the All of Us Research Program. As the dataset was fully deidentified and investigators had no access to information permitting reidentification of participants, the present study was determined not to be human subjects research according to federal guidelines.

Consistent with previous studies,^{7,8} we included participants aged 50 years and older at cohort entry (Figure 1). Cohort entry was defined as the date at which the participant reached 50 years of age or the start of their EHR observation window, whichever occurred later. We excluded participants without EHR and genomic data, with MCI or dementia at or before cohort entry, ADHD diagnosed after MCI or dementia onset, missing demographic information, sex at birth other than male or female, and age greater than 95 years at entry or greater than 100 years at censoring, to limit potential selective survival bias at extreme ages. Follow-up continued until the earliest of cognitive impairment diagnosis, death, or the end of the EHR observation window.

Ascertainment of Diagnoses

ADHD and cognitive impairment were ascertained from EHR using validated algorithms²³ requiring multiple diagnoses or medication records on separate occasions. See details in the eMethods in Supplement 1 and in eTables 2 and 3 in Supplement 2.

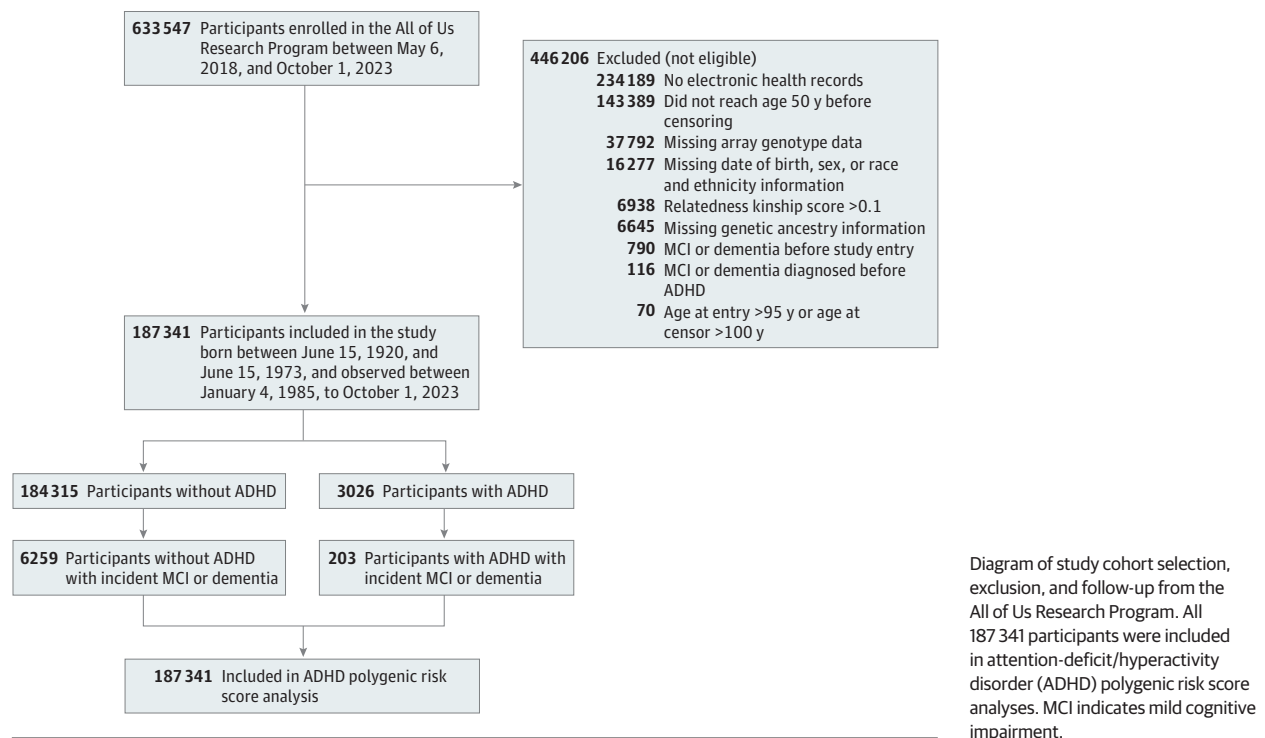
Genomic Data and ADHD PRS Calculation

All of Us Research Program genomic data processing and quality control have been described previously.²¹ We applied additional variant- and sample-level quality control. ADHD PRSs were calculated from the most recent genome-wide association study²⁴ using PRSice version 2.3.5²⁵ under an additive model. Further details are provided in the eMethods in Supplement 1.

Covariates

Covariates included sex at birth (male or female), self-identified race and ethnicity, calendar year of cohort entry, and APOE ε4 carrier status. Sex at birth, race, and ethnicity were self-reported by participants at enrollment using

Figure 1. Study Flow Diagram



response options predefined by the All of Us Research Program.²⁶ Race and ethnicity were included because ADHD diagnosis, cognitive impairment, and health care access can differ across racial and ethnic groups. Participants were grouped as Hispanic or Latino, non-Hispanic Black or African American, non-Hispanic White, or other (non-Hispanic Asian, American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, Middle Eastern or North African, multiple races, and unspecified).

We assessed somatic comorbidities using the Charlson comorbidities (eMethods in [Supplement 1](#)) and summary indices²⁷ derived from EHR data over a 5-year period preceding cohort entry.^{28,29} Neuropsychiatric comorbidities were ascertained from EHR records at any time preceding cohort entry and included depression, anxiety disorder, substance use disorder, bipolar disorder, schizophrenia, and other developmental disorders (eTable 7 in [Supplement 1](#)).

Self-Reported Health and Health Care Utilization

Self-reported health status was assessed using the All of Us Research Program Overall Health survey after MCI or dementia diagnosis (eMethods in [Supplement 1](#)). Health care utilization was measured as the number of emergency department, inpatient, outpatient, and total visits.

Statistical Analysis

Cox proportional hazards models estimated the hazard ratios (HRs) and 95% CIs for the association between ADHD and ADHD PRS and incident MCI or dementia. Models used age as the underlying timescale with delayed entry (left

truncation) to account for varying ages at cohort entry.³⁰ Proportional hazards assumption was assessed using Schoenfeld residuals. Covariate balance across groups was assessed using absolute standardized mean differences (SMDs).³¹

In primary analyses, ADHD was modeled as a time-varying exposure,⁸ meaning pre-entry diagnoses were assigned effect from cohort entry, and postentry diagnoses from the diagnosis date. We fitted additional models with covariate adjustments: (1) adjusted for sex, race, ethnicity, *APOE* $\epsilon 4$ carrier status, and cohort entry year; (2) additionally adjusted for somatic comorbidities; or (3) additionally adjusted for neuropsychiatric comorbidities. For PRS analyses, ADHD PRS was categorized into 3 risk categories: low (quintile 1), intermediate (quintiles 2-4), and high (quintile 5).³² Models followed the same sequential adjustment approach, with race and ethnicity omitted because genetic ancestry was accounted for by residualizing the ADHD PRS. Sensitivity analyses assessed robustness of findings (eMethods in [Supplement 1](#)).

For self-reported health and health care utilization outcomes among participants who developed MCI or dementia, logistic regression models estimated ORs for self-reported health status (suboptimal vs adequate), and negative binomial regression models estimated incidence rate ratios for health care utilization (eMethods in [Supplement 1](#)).

Statistical tests were 2-sided, with $P < .05$ considered significant. Analyses were conducted using R version 4.3.0 (R Foundation) in the All of Us Research Program Researcher Workbench. Data were analyzed from October 2025 to July 2026.

Results

A total of 187 341 participants (mean [SD] age at cohort entry, 56.41 [7.88] years; 107 266 [57%] female and 80 075 [43%] male) were included in our retrospective cohort analysis (Figure 1; Table 1). Participants were followed up for a mean (SD) of 10.26 (6.75) years, with a maximum of 38.73 years. Age at first available EHR record varied substantially across participants (eTable 5 in Supplement 1). The cohort was racially and ethnically diverse, including Hispanic or Latino (25 429 [13.6%]), non-Hispanic Black or African American (32 474 [17.3%]), non-Hispanic White (115 320 [61.6%]), and other (14 118 [7.5%]) participants. The overall prevalence of diagnosed ADHD was 1.6% (3026 individuals). The mean (SD) age at first recorded ADHD diagnosis, which does not necessarily correspond to the age at first diagnosis, was 53.30 (9.71) years (eTable 6 in Supplement 1). Individuals with ADHD entered the cohort at a younger age than those without and had a higher burden of somatic and neuropsychiatric comorbid conditions. Participants stratified by ADHD PRS showed similar values across most characteristics. Notable exceptions were the prevalence of several neuropsychiatric comorbid conditions, including substance use disorder (|SMD|, 0.15), depression (|SMD|, 0.11), and anxiety disorder (|SMD|, 0.11), all of which showed a graded increase from the lowest to the highest PRS category (Table 1). *APOE* ϵ 4 carrier status was similar across all groups, regardless of ADHD diagnosis or ADHD PRS.

ADHD and Incident Cognitive Impairment

The mean follow-up duration was similar among individuals with and without ADHD (|SMD|, 0.05) (Table 1), as well as across ADHD PRS categories (|SMD|, 0.11). During follow-up, 6462 participants (3.4%) received a diagnosis of cognitive impairment (MCI or dementia). Among participants with ADHD, 203 (6.7%) developed cognitive impairment vs 6259 (3.4%) among those without ADHD. Incidence rates of cognitive impairment were 8.96 and 3.29 per 1000 person-years among participants with and without ADHD, respectively. Cumulative incidence was higher in participants with ADHD, with divergence emerging early and widening with age (Figure 2A). A similar pattern was observed across ADHD PRS categories, with incidence rates of 3.04 and 3.67 per 1000 person-years in the low and high categories, respectively (Figure 2B).

In adjusted Cox models, ADHD was associated with a more than 4-fold higher hazard of developing MCI or dementia (HR, 4.60; 95% CI, 3.99-5.29; $P < .001$) (Figure 3A). Further adjustment for somatic comorbidities marginally attenuated the association (HR, 4.04; 95% CI, 3.50-4.66; $P < .001$), whereas adjustment for neuropsychiatric comorbidities yielded a more conservative but still substantial estimate (HR, 3.19; 95% CI, 2.76-3.68; $P < .001$). In subgroup analyses, this association was consistent across strata of sex (male: HR, 4.05; 95% CI, 3.29-4.99; female: HR, 5.15; 95% CI, 4.25-6.23; interaction $P = .15$), race and ethnicity (non-Hispanic White: HR, 4.83; 95% CI, 4.15-5.62; non-Hispanic Black or African American: HR, 6.12; 95% CI, 3.37-11.11; Hispanic: HR, 2.48; 95% CI, 1.11-5.55; other: HR, 2.80; 95% CI, 1.44-5.43; interaction $P = .11$), and *APOE* ϵ 4 sta-

tus (carriers: HR, 3.98; 95% CI, 3.02-5.23; noncarriers: HR, 4.84; 95% CI, 4.11-5.70; interaction $P = .13$).

Higher ADHD PRS was also associated with an increased hazard of MCI or dementia. Participants in the high ADHD PRS category showed a 35% higher hazard compared with those in the low category (adjusted HR, 1.35; 95% CI, 1.25-1.46; $P < .001$), and this estimate remained stable after adjustment for somatic or neuropsychiatric comorbidities (Figure 3B). This association was consistent across strata of sex (male: HR, 1.40; 95% CI, 1.25-1.56; female: HR, 1.32; 95% CI, 1.18-1.47; interaction $P = .41$) and *APOE* ϵ 4 status (carriers: HR, 1.38; 95% CI, 1.20-1.58; noncarriers: HR, 1.34; 95% CI, 1.22-1.47; interaction $P = .85$). In genetic ancestry-stratified analyses, statistically significant associations were observed among participants of European (HR, 1.38; 95% CI, 1.26-1.50; $P < .001$) and African (HR, 1.29; 95% CI, 1.03-1.61; $P = .02$) ancestry.

Sensitivity Analyses

Results were consistent when treating ADHD as a fixed exposure from cohort entry or restricting to ADHD diagnoses before entry only, restricting to individuals entering the cohort at age of 60 years or older, excluding MCI cases, excluding individuals with any neuropsychiatric comorbidities before cohort entry, or restricting the outcome to MCI only (ADHD HR, 3.83; 95% CI, 3.27-4.47; ADHD PRS HR, 1.21; 95% CI, 1.11-1.33; all $P < .001$) or dementia only (ADHD HR, 3.07; 95% CI, 2.12-4.44; ADHD PRS HR, 1.62, 95% CI, 1.40-1.89; all $P < .001$) (eTable 9 in Supplement 1). In a 1:2 matched cohort in which ADHD and comparison individuals were balanced on age, sex, race, ethnicity, cohort entry year, *APOE* ϵ 4 status, somatic and neuropsychiatric comorbidities, and health care utilization, the association was attenuated but remained significant (ADHD HR, 2.26; 95% CI, 1.75-2.92; $P < .001$). Competing risk models accounting for death yielded similar results (ADHD subdistribution HR [sHR], 4.30; 95% CI, 3.73-4.97; $P < .001$; ADHD PRS sHR, 1.19; 95% CI, 1.12-1.26; $P < .001$). A fully adjusted model incorporating sex, race, ethnicity, cohort entry year, *APOE* ϵ 4 status, somatic and neuropsychiatric comorbidities, health care utilization, educational attainment, and lifetime smoking history yielded consistent results (ADHD HR, 3.08; 95% CI, 2.66-3.58; ADHD PRS HR, 1.30; 95% CI, 1.21-1.41; all $P < .001$).

To further address potential reverse causation, we required progressively longer minimum intervals (1 to 10 years) between ADHD diagnosis and cognitive impairment onset (eTable 10 in Supplement 1). The association attenuated as the interval lengthened but remained significant at intervals up to 8 years (HR, 1.44; 95% CI, 1.12-1.86; $P = .005$). It became non-significant at 9 years (HR, 1.28; 95% CI, 0.98-1.68; $P = .07$), by which point 150 ADHD-cognitive impairment cases (74%) were reclassified, reducing statistical power.

Self-Reported Health and Health Care Utilization

Among participants who developed cognitive impairment ($n = 6462$ for utilization outcomes; approximately 4180 for survey outcomes), comorbid ADHD was significantly associated with worse outcomes across multiple domains (Table 2). ADHD was associated with significantly higher odds of suboptimal self-reported mental health (OR, 1.69; 95% CI,

Table 1. Characteristics of the Study Cohort

Characteristic	Overall (N = 187 341)	ADHD diagnosis		SMD	ADHD PRS			SMD ^a
		No ADHD (n = 184 315)	ADHD (n = 3026)		Low (n = 37 469)	Intermediate (n = 112 404)	High (n = 37 468)	
Incident MCI or dementia, No. (%)	6462 (3.4)	6259 (3.4)	203 (6.7)	0.15	1219 (3.3)	3871 (3.4)	1372 (3.7)	0.02
Age at cohort entry, mean (SD), y	56.41 (7.88)	56.47 (7.90)	52.49 (4.93)	0.61	56.83 (8.13)	56.36 (7.84)	56.13 (7.72)	0.09
Follow-up duration, mean (SD), y	10.26 (6.75)	10.26 (6.74)	10.59 (6.80)	0.05	10.72 (6.94)	10.21 (6.72)	9.99 (6.59)	0.11
Sex, No. (%)								
Male	80 075 (42.7)	78 915 (42.8)	1160 (38.3)	0.09	16 636 (44.4)	48 014 (42.7)	15 425 (41.2)	0.07
Female	107 266 (57.3)	105 400 (57.2)	1866 (61.7)		20 833 (55.6)	64 390 (57.3)	22 043 (58.8)	
Race and ethnicity, No. (%)								
Hispanic	25 429 (13.6)	25 236 (13.7)	193 (6.4)	0.52	4281 (11.4)	15 594 (13.9)	5554 (14.8)	0.17
Non-Hispanic Black or African American	32 474 (17.3)	32 322 (17.5)	152 (5.0)		5628 (15.0)	21 576 (19.2)	5270 (14.1)	
Non-Hispanic White	115 320 (61.6)	112 856 (61.2)	2464 (81.4)		25 228 (67.3)	66 447 (59.1)	23 645 (63.1)	
Other ^b	14 118 (7.5)	13 901 (7.5)	217 (7.2)		2332 (6.2)	8787 (7.8)	2999 (8.0)	
APOE ε4 status, No. (%)								
Noncarrier	137 017 (73.1)	134 728 (73.1)	2289 (75.6)	0.06	27 428 (73.2)	81 882 (72.8)	27 707 (73.9)	0.02
Carrier	50 324 (26.9)	49 587 (26.9)	737 (24.4)		10 041 (26.8)	30 522 (27.2)	9761 (26.1)	
Education attainment, No. (%)								
No college degree	95 702 (51.1)	94 377 (51.2)	1325 (43.8)	0.17	16 038 (42.8)	58 041 (51.6)	21 623 (57.7)	0.31
College degree or higher	87 677 (46.8)	86 015 (46.7)	1662 (54.9)		20 798 (55.5)	51 864 (46.1)	15 015 (40.1)	
Unknown, refused to answer or missing	3962 (2.1)	3923 (2.1)	39 (1.3)		633 (1.7)	2499 (2.2)	830 (2.2)	
Smoked ≥100 cigarettes in lifetime, No. (%)								
Yes	82 103 (43.8)	80 626 (43.7)	1477 (48.8)	0.1	14 480 (38.6)	49 205 (43.8)	18 418 (49.2)	0.22
No	100 946 (53.9)	99 468 (54.0)	1478 (48.8)		22 164 (59.2)	60 623 (53.9)	18 159 (48.5)	
Unknown, refused to answer or missing	4292 (2.3)	4221 (2.3)	71 (2.3)		825 (2.2)	2576 (2.3)	891 (2.4)	
Charlson Comorbidity Index, mean (SD)	0.25 (0.93)	0.24 (0.92)	0.57 (1.35)	0.28	0.21 (0.85)	0.25 (0.93)	0.29 (1.01)	0.09
Myocardial infarction, No. (%)	1233 (0.7)	1194 (0.6)	39 (1.3)	0.07	188 (0.5)	747 (0.7)	298 (0.8)	0.04
Congestive heart failure, No. (%)	1955 (1.0)	1908 (1.0)	47 (1.6)	0.05	342 (0.9)	1177 (1.0)	436 (1.2)	0.02
Peripheral vascular disease, No. (%)	1458 (0.8)	1406 (0.8)	52 (1.7)	0.09	239 (0.6)	877 (0.8)	342 (0.9)	0.03
Cerebrovascular disease, No. (%)	1901 (1.0)	1827 (1.0)	74 (2.4)	0.11	296 (0.8)	1176 (1.0)	429 (1.1)	0.04
Chronic pulmonary disease, No. (%)	10 452 (5.6)	9963 (5.4)	489 (16.2)	0.35	1702 (4.5)	6267 (5.6)	2483 (6.6)	0.09
Rheumatic disease, No. (%)	2067 (1.1)	1990 (1.1)	77 (2.5)	0.11	366 (1.0)	1239 (1.1)	462 (1.2)	0.02
Peptic ulcer disease, No. (%)	910 (0.5)	868 (0.5)	42 (1.4)	0.1	152 (0.4)	535 (0.5)	223 (0.6)	0.03
Liver disease, No. (%)	4499 (2.4)	4306 (2.3)	193 (6.4)	0.19	725 (1.9)	2668 (2.4)	1106 (3.0)	0.06
Diabetes without complications, No. (%)	6242 (3.3)	6063 (3.3)	179 (5.9)	0.13	1013 (2.7)	3761 (3.3)	1468 (3.9)	0.07
Diabetes with complications, No. (%)	2754 (1.5)	2677 (1.5)	77 (2.5)	0.08	440 (1.2)	1689 (1.5)	625 (1.7)	0.04
Hemiplegia or paraplegia, No. (%)	563 (0.3)	Not reported ^c	Not reported ^c	NA	104 (0.3)	317 (0.3)	142 (0.4)	0.02
Kidney disease, No. (%)	2121 (1.1)	2069 (1.1)	52 (1.7)	0.05	393 (1.0)	1243 (1.1)	485 (1.3)	0.02
Cancer, No. (%)	3174 (1.7)	3065 (1.7)	109 (3.6)	0.12	622 (1.7)	1894 (1.7)	658 (1.8)	0.01
Metastatic solid tumor, No. (%)	703 (0.4)	682 (0.4)	21 (0.7)	0.04	118 (0.3)	418 (0.4)	167 (0.4)	0.02
AIDS/HIV, No. (%)	1015 (0.5)	980 (0.5)	35 (1.2)	0.07	144 (0.4)	626 (0.6)	245 (0.7)	0.04
Head injuries, No. (%)	1147 (0.6)	1074 (0.6)	73 (2.4)	0.15	222 (0.6)	685 (0.6)	240 (0.6)	0.01
Depression, No. (%)	15 179 (8.1)	14 186 (7.7)	993 (32.8)	0.66	2491 (6.6)	9042 (8.0)	3646 (9.7)	0.11
Anxiety disorder, No. (%)	12 858 (6.9)	11 994 (6.5)	864 (28.6)	0.61	2077 (5.5)	7669 (6.8)	3112 (8.3)	0.11
Substance use disorder, No. (%)	11 944 (6.4)	11 415 (6.2)	529 (17.5)	0.35	1661 (4.4)	7233 (6.4)	3050 (8.1)	0.15

(continued)

Table 1. Characteristics of the Study Cohort (continued)

Characteristic	Overall (N = 187 341)	ADHD diagnosis		[SMD]	ADHD PRS			[SMD] ^a
		No ADHD (n = 184 315)	ADHD (n = 3026)		Low (n = 37 469)	Intermediate (n = 112 404)	High (n = 37 468)	
Bipolar disorder, No. (%)	3123 (1.7)	2843 (1.5)	280 (9.3)	0.35	413 (1.1)	1853 (1.6)	857 (2.3)	0.09
Schizophrenia, No. (%)	1541 (0.8)	1467 (0.8)	74 (2.4)	0.13	242 (0.6)	933 (0.8)	366 (1.0)	0.04
Other developmental disorders, No. (%)	310 (0.2)	270 (0.1)	40 (1.3)	0.14	53 (0.1)	189 (0.2)	68 (0.2)	0.01
Health care visits (1 y prior cohort entry), mean (SD)	2.83 (9.17)	2.72 (8.93)	9.13 (17.60)	0.46	2.53 (8.46)	2.85 (9.19)	3.06 (9.78)	0.06

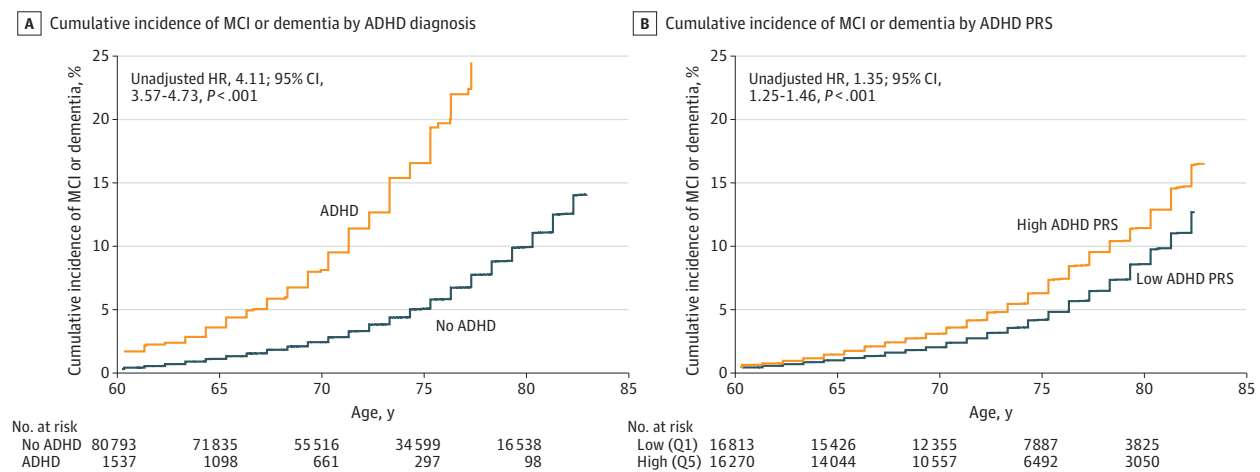
Abbreviations: ADHD, attention-deficit/hyperactivity disorder; APOE, apolipoprotein E; MCI, mild cognitive impairment; NA, not applicable; SMD, standardized mean difference.

^a For ADHD PRS, [SMD] are computed comparing high (quintile 5) vs low (quintile 1) categories.

^b Includes non-Hispanic Asian, American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, Middle Eastern or North African, multiple races, and unspecified.

^c Per the All of Us Data and Statistics Dissemination Policy, participant counts of 20 or fewer, and values that would enable their derivation, are not reported.

Figure 2. Cumulative Incidence Curves Showing Mild Cognitive Impairment (MCI) or Dementia by Attention-Deficit/Hyperactivity Disorder (ADHD) Diagnosis and ADHD Polygenic Risk Score (PRS)



A, Cumulative incidence of MCI or dementia by ADHD diagnosis (ADHD vs no ADHD). B, Cumulative incidence of MCI or dementia by ADHD PRS (low [quintile 1; Q1] vs high [quintile 5; Q5]). Hazard ratios (HRs) and P values from unadjusted Cox proportional hazards models.

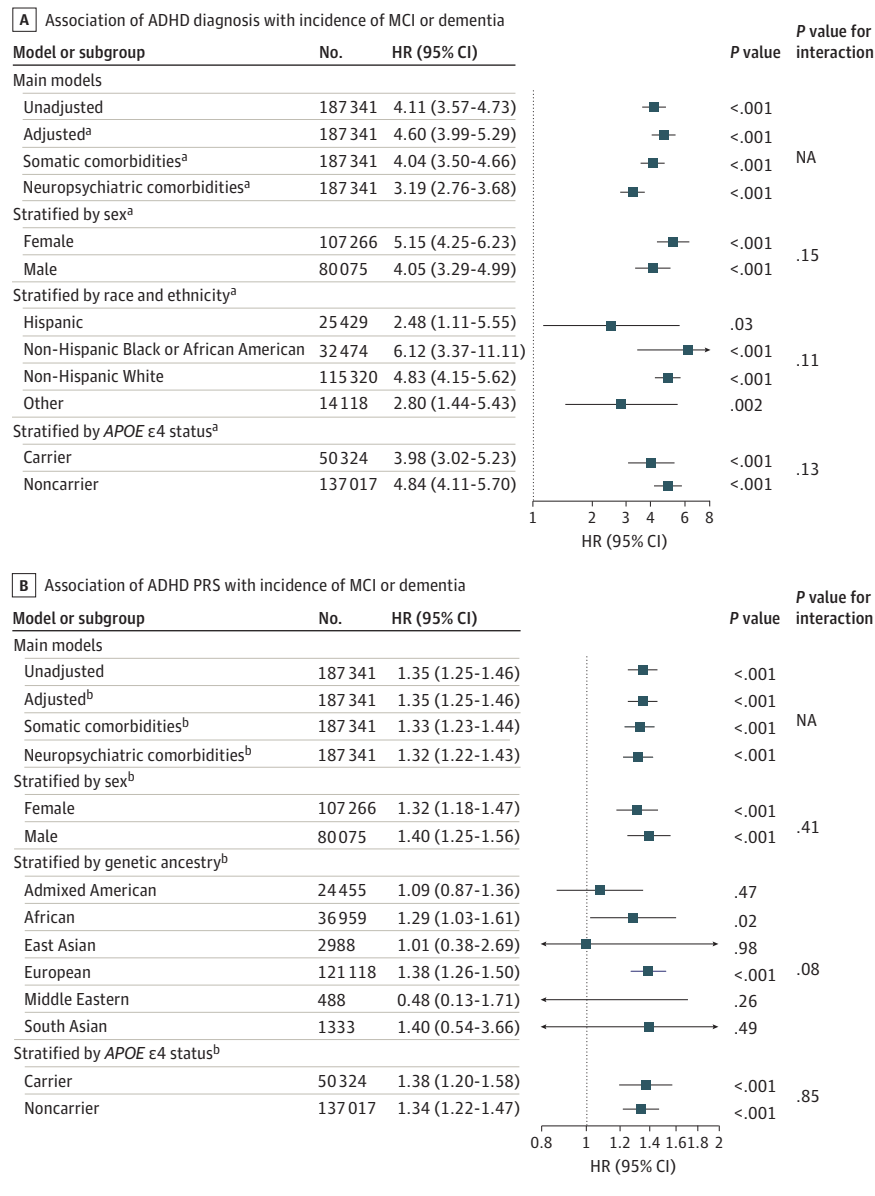
1.18-2.39; adjusted $P = .01$), quality of life (OR, 1.66; 95% CI, 1.14-2.39; adjusted $P = .02$), and social satisfaction (OR, 2.39; 95% CI, 1.69-3.37; adjusted $P < .001$). No significant differences were observed for general or physical health. In the year following cognitive impairment diagnosis, ADHD was associated with 34% higher outpatient visit rates (incidence rate ratio [IRR], 1.34; 95% CI, 1.15-1.56; adjusted $P = .001$) and 20% higher total health care utilization (IRR, 1.20; 95% CI, 1.09-1.34; adjusted $P = .003$), even after adjustment for prediagnosis visit rates. Emergency department and inpatient visits did not differ significantly by ADHD diagnosis. Conversely, higher ADHD PRS was associated with higher emergency department visit rates (IRR, 1.42; 95% CI, 1.15-1.75; adjusted $P = .005$).

Discussion

This cohort study evaluated the association between ADHD and incident cognitive impairment in a large, diverse sample from the All of Us Research Program, addressing key limitations

from prior research by complementing EHR-based ADHD ascertainment with independent genetic evidence (ADHD PRS), adjusting for a broader range of confounders (including APOE $\epsilon 4$ carrier status and health care utilization), and examining associated mental health and health care outcomes among older adults with MCI or dementia and comorbid ADHD. ADHD diagnosis was associated with a higher risk of subsequent cognitive impairment, and this association remained across multiple sensitivity analyses. Analyses leveraging genomic data from the same study population provided convergent evidence, with higher ADHD PRS also associated with increased risk of cognitive impairment. Finally, among individuals with cognitive impairment, comorbid ADHD was associated with worse self-reported mental health, quality of life, social satisfaction, and higher health care utilization.

Our findings align with and expand a growing literature supporting an association between ADHD and increased risk of MCI and dementia. Studies using population-based health claims databases from Taiwan,⁵ Sweden,⁷ Israel,⁸ and Germany⁹ have reported elevated risk estimates that are

Figure 3. Forest Plots Showing Association of Attention-Deficit/Hyperactivity Disorder (ADHD) Diagnosis and ADHD Polygenic Risk Score (PRS) With Incident Mild Cognitive Impairment (MCI) or Dementia

A, Hazard ratios (HRs) for the association between ADHD diagnosis and incident MCI or dementia. B, HRs comparing high vs low ADHD PRS categories and incident MCI or dementia. Panels A and B display main models with sequential covariate adjustment, followed by subgroup analyses stratified by sex, race/ethnicity (panel A) or genetic ancestry (panel B), and APOE ε4 carrier status. In subgroup analyses, the stratification variable was not included in model adjustment. For panel B, only the high vs low ADHD PRS comparison is shown for illustrative purposes; statistical models were fitted using all participants across the 3 ADHD PRS categories (low, intermediate, and high).

^aAdjusted for sex, race, ethnicity, APOE ε4 carrier status, and cohort entry year.

^bAdjusted for sex, APOE ε4 carrier status, and cohort entry year. Genetic ancestry principal components are accounted for in PRS residualization.

broadly comparable in magnitude to ours, indicating that the association is not confined to a single health care setting or diagnostic context. Relative to recent case-control work in the All of Us Research Program,¹² we applied a retrospective cohort design with time-to-event analyses, a more stringent diagnostic ascertainment algorithm, and we leveraged genomic data. Studies based solely on EHR for diagnostic ascertainment are susceptible to biases related to health care access, clinician recognition, and changing diagnostic practices over time.¹⁶ Within the same population, the convergence of EHR-based and ADHD PRS findings, 2 independent sources of evidence of ADHD, strengthens confidence in the association.

Notably, the association persisted despite adjustment for variables often considered potential sources of confounding

or partial mediation in studies of ADHD and cognitive outcomes, including educational attainment, smoking history, and psychiatric and somatic comorbidities.^{33,34} Adjustment for psychiatric comorbidities attenuated but did not eliminate the association, consistent with prior literature,^{5,7,12} and the ADHD PRS association likewise remained stable. It also persisted in a propensity score-matched cohort balanced on each individual neuropsychiatric condition and in a sensitivity analysis restricted to participants without any history of neuropsychiatric comorbidity at cohort entry. Estimates attenuated progressively as more confounders were addressed, with the matched analyses yielding the most conservative values, likely through more complete balancing of confounders, including health care utilization, than covariate adjustment alone. That a substantial association persisted under this most

Table 2. Association Between Attention-Deficit/Hyperactivity Disorder (ADHD) and Self-Reported Health Status and Health Care Resource Utilization After Mild Cognitive Impairment (MCI) or Dementia Diagnosis

	ADHD diagnosis ^a		ADHD-PRS ^a		
Outcome	OR/IRR (95% CI) ^b	Adjusted <i>P</i> value ^c	OR/IRR (95% CI) ^b	Adjusted <i>P</i> value ^c	No.
Suboptimal self-reported health status ^d					
General health	1.05 (0.73-1.49)	.84	1.11 (0.91-1.36)	.39	4185
Mental health	1.69 (1.18-2.39)	.01	1.14 (0.91-1.42)	.35	4182
Physical health	1.31 (0.93-1.85)	.20	1.27 (1.04-1.56)	.05	4171
Quality of life	1.66 (1.14-2.39)	.02	1.30 (1.02-1.66)	.07	4167
Social satisfaction	2.39 (1.69-3.37)	<.001	1.26 (1.00-1.60)	.11	4179
Health care utilization 1 y after diagnosis ^e					
ED visits	0.91 (0.63-1.33)	.67	1.42 (1.15-1.75)	.005	6462
Inpatient visits	1.34 (0.93-1.99)	.20	0.99 (0.80-1.22)	.90	6462
Outpatient visits	1.34 (1.15-1.56)	.001	1.06 (0.98-1.16)	.22	6462
Total visits	1.20 (1.09-1.34)	.003	1.02 (0.96-1.08)	.67	6462

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; ED, emergency department; IRR, incidence rate ratio; MCI, mild cognitive impairment; OR, odds ratio.

^a All models adjusted for age, sex, race, ethnicity, Charlson Comorbidity Index, and presence of any neuropsychiatric comorbidity. In ADHD PRS, race and ethnicity were omitted because genetic ancestry principal components were accounted for in PRS residualization. For ADHD PRS, only the high vs low risk comparison is shown for illustrative purposes; statistical models were fitted using all participants across the 3 categories (low, intermediate, and high).

^b ORs from logistic regression models are reported for suboptimal self-reported

health status outcomes. IRRs from negative binomial regression is reported for health care utilization and medication burden outcomes.

^c P values adjusted for false discovery rate using the Benjamini-Hochberg procedure across all reported outcomes. Significance level: <.05.

^d Models were additionally adjusted for time from MCI or dementia diagnosis to survey date.

^e Models were additionally adjusted for total visits 1 year prior to MCI or dementia diagnosis.

stringent approach argues against confounding as a full explanation, though residual confounding or mediation cannot be excluded.

No clear effect modification by sex or race and ethnicity was observed, though caution is warranted given demographic differences in ADHD recognition.^{35,36} The association between ADHD PRS and cognitive decline varied considerably by genetic ancestry, which is not unexpected given that the ADHD PRS was calculated using summary statistics from a genome-wide association study composed predominantly of individuals of European ancestry.²⁴ We observed no evidence that the association with ADHD differed by *APOE* $\epsilon 4$ status, which is the major genetic risk factor for Alzheimer disease.³⁷ These findings suggest that ADHD liability contributes to cognitive impairment through pathways that are at least partially distinct from *APOE* $\epsilon 4$ -driven risk, consistent with prior work.¹⁵

A novel finding is that among individuals who developed cognitive impairment, a comorbid ADHD diagnosis was associated with worse mental health, lower social satisfaction, and greater health care utilization, whereas ADHD PRS showed more limited associations with postdiagnosis outcomes, except for higher emergency department visit rates. Although psychiatric comorbidities broadly worsen quality of life and care burden in dementia,^{38,39} the specific contribution of ADHD comorbidity to outcomes after cognitive impairment diagnosis has not been examined. The divergence between clinical diagnosis and genetic liability for these outcomes suggests that the full phenotypic expression of ADHD, including its life-long comorbidities and social outcomes, contributes to post-

diagnosis burden in ways that genetic liability alone does not fully capture.

The mechanisms linking ADHD to cognitive decline remain unknown and likely involve multiple interacting processes. One hypothesis is that the ADHD-related liability reflects, in part, reduced cognitive resilience and reserve,^{15,40} potentially arising from neurodevelopmental influences on attention and executive function networks.⁴¹ In this framework, lower average baseline cognitive performance observed among individuals with ADHD⁴² would increase susceptibility to crossing the clinical threshold for MCI or dementia as neuropathology accumulates. A second hypothesis is that ADHD increases exposure to cumulative risk factors for cognitive impairment, including chronic stress, social adversity, sleep disturbance, physical inactivity, poor diet quality, cardiometabolic risk, injuries, and substance use.^{43,44} These pathways are difficult to quantify precisely in retrospective analyses, and disentangling their relative contribution will require longitudinal studies with repeated measurement. However, existing cohorts of individuals with ADHD diagnosed in childhood are only now approaching ages when dementia outcomes can be studied directly.⁴⁵ The ADHD-PRS association should also be interpreted cautiously, as it may reflect liability to ADHD and its life-course correlates, shared liability with other psychiatric or neurodevelopmental traits, or broader pleiotropic pathways rather than a direct causal effect on dementia neuropathology.⁴⁶

An association between ADHD and cognitive impairment could have meaningful clinical implications, particularly as rec-

ognition of ADHD across the lifespan increases and more adults receive diagnoses later in life. A key unanswered question is whether ADHD treatment modifies risk for cognitive impairment. A prior report suggested no elevated dementia risk among older adults with late-life ADHD diagnosis who were being treated with stimulants, though confounding cannot be excluded.⁸ We could not address treatment effects here due to limited data on lifetime treatment for ADHD. Notably, among individuals with cognitive impairment, comorbid ADHD was associated with greater support needs, possibly reflecting additive impacts of lifelong executive dysfunction and emerging cognitive decline, underscoring the importance of proactive monitoring and tailored interventions in this population.

Limitations

Findings from this study should be interpreted considering some limitations. Several features limit causal interpretation. Because this is an observational study, residual confounding cannot be fully excluded. Reverse causation is a particular concern, given the long prodromal phase of dementia⁴⁷; indeed, in lagged analyses, the association attenuated as longer intervals between ADHD diagnosis and cognitive impairment were required, and a contribution from prodromal misattribution cannot be excluded. Differential health care contact could also promote greater ascertainment of cognitive impairment among individuals with ADHD; the association nonetheless persisted after adjustment for and matching on health care utilization. The ADHD-PRS association provides complementary evidence not subject to reverse causation from prodromal cognitive decline, although it does not establish causality as it may reflect shared liability or other pleiotropic pathways. EHR diagnoses may be underascertained or misclassified, limiting identification of ADHD, cognitive impairment, and comorbidities. For most participants,

available EHR observation began in midlife or later. Thus, age at first recorded ADHD diagnosis reflects the first ADHD documentation available in the EHR, not necessarily ADHD onset or first clinical diagnosis. ADHD PRS was derived from summary statistics of a genome-wide association study predominantly composed of individuals of European ancestry,²⁴ limiting its generalizability to other ancestry groups. We could not adequately evaluate the role of ADHD pharmacologic and non-pharmacological treatment in modifying risk, and this remains an important open question for future studies. Selection bias that occurs in volunteer research cohorts like this one may limit generalizability, and the population may not represent the individuals with ADHD commonly found in clinical care. For example, most of the individuals with ADHD were women, contrasting with the broader literature, where ADHD is roughly comparable by sex in adulthood.⁴⁸ Similarly, individuals with ADHD had higher educational attainment. Although this finding contrasts with the broader literature linking ADHD to lower educational attainment,⁴⁴ prior studies in older populations have reported no differences in education or income between individuals with and without ADHD.^{49,50}

Conclusions

In summary, in this cohort study, ADHD was associated with increased risk of incident MCI or dementia, and this association was observed using both EHR-based diagnoses and complementary genetic liability measures using ADHD PRS. These findings support ADHD as a relevant factor in later-life cognitive health and call for clinical guidelines recommending cognitive monitoring in adults with ADHD, while recognizing that the mechanisms and causal nature of this association remain to be established.

ARTICLE INFORMATION

Accepted for Publication: August 10, 2026.

Published Online: October 7, 2026.

doi:10.1001/jamapsychiatry.2026.3165

Open Access: This is an open access article distributed under the terms of the [CC-BY-NC-ND License](#), which does not permit alteration or commercial use, including those for text and data mining, AI training, and similar technologies. © 2026 Bauer-Negrini G et al. *JAMA Psychiatry*.

Author Affiliations: Department of Psychiatry, University of Pittsburgh, Pittsburgh, Pennsylvania (Bauer-Negrini, Leffa, Ferreira, Bellaver, Povala, Medeiros, Lussier, Tudorascu, Pascoal); Developmental Evidence Synthesis, Prediction, and Implementation Lab, Centre for Innovation in Mental Health, School of Psychology, Faculty of Environmental and Life Sciences, University of Southampton, Southampton, United Kingdom (Cortese); Clinical and Experimental Sciences, Faculty of Medicine, University of Southampton, Southampton, United Kingdom (Cortese); Hampshire and Isle of Wight Healthcare National Health Service Foundation Trust, Southampton, United Kingdom (Cortese); Hassenfeld Children's Hospital at New York University Langone, New York University Child Study Center, New York (Cortese);

DiMePre-J-Department of Precision and Regenerative Medicine-Jonic Area, University of Bari "Aldo Moro," Bari, Italy (Cortese); ADHD Outpatient Program & Development Psychiatry Program, Hospital de Clínicas de Porto Alegre, Porto Alegre, Brazil (Rohde); National Institute of Developmental Psychiatry & National Center for Innovation and Research in Mental Health, São Paulo, Brazil (Rohde); Medical Council, Centro Universitário Max Planck (UniMAX), Indaiatuba, Brazil (Rohde); Departments of Psychiatry, Psychology, Pediatrics, Clinical and Translational Science, University of Pittsburgh, Pittsburgh, Pennsylvania (Molina); Department of Neurology, University of Pittsburgh, Pittsburgh, Pennsylvania (Pascoal).

Author Contributions: Drs Bauer-Negrini and Pascoal had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Bauer-Negrini and Leffa are co-first authors. **Concept and design:** Bauer-Negrini, Leffa, Ferreira, Rohde, Molina, Pascoal. **Acquisition, analysis, or interpretation of data:** Bauer-Negrini, Leffa, Bellaver, Povala, Medeiros, Lussier, Tudorascu, Cortese, Rohde, Pascoal. **Drafting of the manuscript:** Bauer-Negrini, Leffa, Rohde, Pascoal.

Critical review of the manuscript for important intellectual content: All authors.

Statistical analysis: Bauer-Negrini, Leffa, Povala, Tudorascu, Rohde.

Obtained funding: Pascoal.

Administrative, technical, or material support: Bellaver, Medeiros, Lussier.

Supervision: Cortese, Rohde, Molina, Pascoal.

Conflict of Interest Disclosures: Dr Pascoal reported serving on a Data Safety Monitoring Board for a National Institutes of Health award based at the University of Cincinnati and previously serving on a scientific advisory board for Johnson & Johnson. Drs Bauer-Negrini and Povala reported grants from the Alzheimer's Association during the conduct of the study. Dr Tudorascu reported a grant from the University of Pittsburgh, paid to the University of Pittsburgh, during the conduct of the study and an honorarium from the American Psychiatric Association for a webinar outside the submitted work. Dr Leffa reported travel awards from the American Professional Society of ADHD and Related Disorders to attend its 2024, 2025, and 2026 annual meetings; a travel award to attend the 10th World Congress on attention-deficit/hyperactivity disorder; and reimbursement for a lecture at Congresso Aprender Criança 2026, outside the submitted work. Dr Cortese reported

grants from the National Institute for Health and Care Research and the European Research Agency, honoraria from Medice, and reimbursement for travel and accommodation related to lectures and educational activities for the Association for Child and Adolescent Mental Health, Canadian ADHD Resource Alliance, British Association for Psychopharmacology, Healthcare Convention, and CCM Group. Dr Rohde reported grant or research support, consulting fees, and speakers' bureau fees from Abdi Ibrahim, Abbott, Aché, Adium, Apsen, Bial, Cellera, EMS, Hypera Pharma, Knight Therapeutics, Libbs, Medice, Novartis/Sandoz, Pfizer/Upjohn/Viatris, Shire/Takeda, and Torrent; unrestricted educational and research support to the attention-deficit/hyperactivity disorder and Juvenile Bipolar Disorder Outpatient Programs that he chairs from Novartis/Sandoz and Shire/Takeda; and authorship royalties from ArtMed and Oxford University Press. No other disclosures were reported.

Funding/Support: The All of Us Research Program is supported by the National Institutes of Health, Office of the Director, through the following awards and agreements: Regional Medical Centers, 1 OT2 OD026549, 1 OT2 OD026554, 1 OT2 OD026557, 1 OT2 OD026556, 1 OT2 OD026550, 1 OT2 OD026552, 1 OT2 OD026553, 1 OT2 OD026548, 1 OT2 OD026551, and 1 OT2 OD026555; Interagency Agreement (IAA), AOD 16037; Federally Qualified Health Centers, HHSN 263201600085U; Data and Research Center, 5 U2C OD023196; Biobank, 1 U24 OD023121; Participant Center, U24 OD023176; Participant Technology Systems Center, 1 U24 OD023163; Communications and Engagement, 3 OT2 OD023205 and 3 OT2 OD023206; and Community Partners, 1 OT2 OD025277, 3 OT2 OD025315, 1 OT2 OD025337, and 1 OT2 OD025276. Dr Pascoal is supported by National Institute on Aging awards 5 R01 AG073267 and 5 R01 AG075336. Dr Bauer-Negrini, Dr Ferreira, Dr Bellaver, and Dr Povola are supported by the Alzheimer's Association through awards AARF-D-23-1150249, AARF-D-22-923814, AARF-D-22-974627, and 24AARF-D-1243899, respectively. Dr Cortese, a National Institute for Health and Care Research (NIHR) Research Professor (NIHR303122), is funded by the NIHR for this research project and is also supported by NIHR grants NIHR203684, NIHR203035, NIHR130077, NIHR128472, and RP-PG-0618-20003, as well as European Research Executive Agency grant 101095568-HORIZONHLTH-2022-DISEASE-07-03.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Disclaimer: The views expressed in this publication are those of the authors and not necessarily those of the National Institute for Health and Care Research, National Health Service, or the UK Department of Health and Social Care.

Data Sharing Statement: See Supplement 3.

Additional Contributions: We thank the National Institutes of Health's All of Us Research Program and its participants, without whom this research would not have been possible.

REFERENCES

1. Leffa DT, Caye A, Rohde LA. ADHD in children and adults: diagnosis and prognosis. *Curr Top Behav Neurosci*. 2022;57:1-18. doi:10.1007/7854_2022_329
2. Song P, Zha M, Yang Q, Zhang Y, Li X, Rudan I; Global Health Epidemiology Reference Group (GHERG). The prevalence of adult attention-deficit hyperactivity disorder: a global systematic review and meta-analysis. *J Glob Health*. 2021;11:04009. doi:10.7189/jogh.11.04009
3. Kooij JJ, Michielsen M, Kruihof H, Bijlenga D. ADHD in old age: a review of the literature and proposal for assessment and treatment. *Expert Rev Neurother*. 2016;16(12):1371-1381. doi:10.1080/14737175.2016.1204914
4. Goodman DW, Cortese S, Faraone SV. Why is ADHD so difficult to diagnose in older adults? *Expert Rev Neurother*. 2024;24(10):941-944. doi:10.1080/14737175.2024.2385932
5. Tzeng NS, Chung CH, Lin FH, et al. Risk of dementia in adults with ADHD: a nationwide, population-based cohort study in Taiwan. *J Atten Disord*. 2019;23(9):995-1006. doi:10.1177/1087054717714057
6. Du Rietz E, Brikkel I, Butwicki A, et al. Mapping phenotypic and aetiological associations between ADHD and physical conditions in adulthood in Sweden: a genetically informed register study. *Lancet Psychiatry*. 2021;8(9):774-783. doi:10.1016/S2215-0366(21)00171-1
7. Dobrosavljevic M, Zhang L, Garcia-Argibay M, et al. Attention-deficit/hyperactivity disorder as a risk factor for dementia and mild cognitive impairment: a population-based register study. *Eur Psychiatry*. 2021;65(1):1-19. doi:10.1192/j.eurpsy.2021.2261
8. Levine SZ, Rotstein A, Kodesh A, et al. Adult attention-deficit/hyperactivity disorder and the risk of dementia. *JAMA Netw Open*. 2023;6(10):e2338088. doi:10.1001/jamanetworkopen.2023.38088
9. Libutzki B, Neukirch B, Reif A, Hartman CA. Somatic burden of attention-deficit/hyperactivity disorder across the lifecourse. *Acta Psychiatr Scand*. 2024;150(2):105-117. doi:10.1111/acps.13694
10. Golimstok Á, Basalo MJG, Majul M, et al. Adult attention deficit-hyperactivity disorder is associated with Lewy body disease and cognitive impairment: a prospective cohort study with 15-year follow-up. *Am J Geriatr Psychiatry*. 2024;32(9):1063-1077. doi:10.1016/j.jagp.2024.04.005
11. Fluegge K, Fluegge K. Antecedent ADHD, dementia, and metabolic dysregulation: a U.S. based cohort analysis. *Neurochem Int*. 2018;112:255-258. doi:10.1016/j.neuint.2017.08.005
12. Rast JE, Rosso AL, James BD, et al. Association of neurodevelopmental conditions with Alzheimer's disease and related dementias and Parkinson's disease. *J Gerontol A Biol Sci Med Sci*. 2025;81(4):glaf281. doi:10.1093/gerona/glaf281
13. Ronald A, de Bode N, Polderman TJC. Systematic review: how the attention-deficit/hyperactivity disorder polygenic risk score adds to our understanding of ADHD and associated traits. *J Am Acad Child Adolesc Psychiatry*. 2021;60(10):1234-1277. doi:10.1016/j.jaac.2021.01.019
14. Furrer RA, Turkheimer EN. Polygenic risk scores are not genetic predispositions. *Nat Hum Behav*. 2026;10(7):1177-1178. doi:10.1038/s41562-026-02475-8
15. Leffa DT, Ferrari-Souza JP, Bellaver B, et al; Alzheimer's Disease Neuroimaging Initiative. Genetic risk for attention-deficit/hyperactivity disorder predicts cognitive decline and development of Alzheimer's disease pathophysiology in cognitively unimpaired older adults. *Mol Psychiatry*. 2023;28(3):1248-1255. doi:10.1038/s41380-022-01867-2
16. Goldstein ND, Kahal D, Testa K, Gracely EJ, Burstyn I. Data quality in electronic health record research: an approach for validation and quantitative bias analysis for imperfectly ascertained health outcomes via diagnostic codes. *Harv Data Sci Rev*. 2022;4(2). doi:10.1162/99608f92.cbe67e91
17. Landes SD, London AS. Self-reported ADHD and adult health in the United States. *J Atten Disord*. 2021;25(1):3-13. doi:10.1177/1087054718757648
18. Biederman J, Faraone SV, Spencer TJ, Mick E, Monuteaux MC, Aleardi M. Functional impairments in adults with self-reports of diagnosed ADHD: a controlled study of 1001 adults in the community. *J Clin Psychiatry*. 2006;67(4):524-540. doi:10.4088/JCP.v67n0403
19. Swensen AR, Birnbaum HG, Secnik K, Marynchenko M, Greenberg P, Claxton A. Attention-deficit/hyperactivity disorder: increased costs for patients and their families. *J Am Acad Child Adolesc Psychiatry*. 2003;42(12):1415-1423. doi:10.1097/00004583-200312000-00008
20. Denny JC, Rutter JL, Goldstein DB, et al; All of Us Research Program Investigators. The "All of Us" research program. *N Engl J Med*. 2019;381(7):668-676. doi:10.1056/NEJMs1809937
21. Bick AG, Metcalf GA, Mayo KR, et al; All of Us Research Program Genomics Investigators. Genomic data in the All of Us Research Program. *Nature*. 2024;627(8003):340-346. doi:10.1038/s41586-023-06957-x
22. von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP; STROBE Initiative. The Strengthening of Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *J Clin Epidemiol*. 2008;61(4):344-349. doi:10.1016/j.jclinepi.2007.11.008
23. Slaby I, Hain HS, Abrams D, et al. An electronic health record (EHR) phenotype algorithm to identify patients with attention deficit hyperactivity disorders (ADHD) and psychiatric comorbidities. *J Neurodev Disord*. 2022;14(1):37. doi:10.1186/s11689-022-09447-9
24. Demontis D, Walters GB, Athanasiadis G, et al; ADHD Working Group of the Psychiatric Genomics Consortium; iPSYCH-Broad Consortium. Genome-wide analyses of ADHD identify 27 risk loci, refine the genetic architecture and implicate several cognitive domains. *Nat Genet*. 2023;55(2):198-208. doi:10.1038/s41588-022-01285-8
25. Euesden J, Lewis CM, O'Reilly PF. PRSice: polygenic risk score software. *Bioinformatics*. 2015;31(9):1466-1468. doi:10.1093/bioinformatics/btu848
26. Cronin RM, Jerome RN, Mapes B, et al; Vanderbilt University Medical Center Pilot Team, and the Participant Provided Information Committee. Development of the initial surveys for the All of Us Research Program. *Epidemiology*.

- 2019;30(4):597-608. doi:10.1097/EDE.0000000000001028
27. Charlson ME, Pompei P, Ales KL, MacKenzie CR. A new method of classifying prognostic comorbidity in longitudinal studies: development and validation. *J Chronic Dis*. 1987;40(5):373-383. doi:10.1016/0021-9681(87)90171-8
 28. Gasparini A. Comorbidity: an R package for computing comorbidity scores. *J Open Source Softw*. 2018;3(23):648. doi:10.21105/joss.00648
 29. Westerberg M, Garma H, Ludvigsson JF, Stattin P, Gedeberg R. Discriminative ability of the Charlson Comorbidity Index for long-term mortality in a general population: nationwide, population-based study of 10 million adults in Sweden. *Clin Epidemiol*. 2025;17:983-993. doi:10.2147/CLEP.S568699
 30. Thiébaud AC, Bénichou J. Choice of time-scale in Cox's model analysis of epidemiologic cohort data: a simulation study. *Stat Med*. 2004;23(24):3803-3820. doi:10.1002/sim.2098
 31. Yang D, Dalton JE. A unified approach to measuring the effect size between two groups using SAS. Published 2012. Accessed September 1, 2026. <https://www.semanticscholar.org/paper/A-unified-approach-to-measuring-the-effect-size-two-Yang-Dalton/6cf4bd36ca4c90006a5d6563f646a391c255581b>
 32. Li JJ. The positive end of the polygenic score distribution for ADHD: a low risk or a protective factor? *Psychol Med*. 2021;51(1):102-111. doi:10.1017/S0033291719003039
 33. Becker S, Chowdhury M, Tavilsup P, Seitz D, Callahan BL. Risk of neurodegenerative disease or dementia in adults with attention-deficit/hyperactivity disorder: a systematic review. *Front Psychiatry*. 2023;14:1158546. doi:10.3389/fpsy.2023.1158546
 34. Becker S, Sharma MJ, Callahan BL. ADHD and neurodegenerative disease risk: a critical examination of the evidence. *Front Aging Neurosci*. 2022;13:826213. doi:10.3389/fnagi.2021.826213
 35. Shi Y, Hunter Guevara LR, Dykhoff HJ, et al. Racial disparities in diagnosis of attention-deficit/hyperactivity disorder in a US national birth cohort. *JAMA Netw Open*. 2021;4(3):e210321. doi:10.1001/jamanetworkopen.2021.0321
 36. Cénat JM, Blais-Rochette C, Morse C, et al. Prevalence and risk factors associated with attention-deficit/hyperactivity disorder among US Black individuals: a systematic review and meta-analysis. *JAMA Psychiatry*. 2021;78(1):21-28. doi:10.1001/jamapsychiatry.2020.2788
 37. Knopman DS, Amieva H, Petersen RC, et al. Alzheimer disease. *Nat Rev Dis Primers*. 2021;7(1):33. doi:10.1038/s41572-021-00269-y
 38. Mo M, Zacarias-Pons L, Hoang MT, et al. Psychiatric disorders before and after dementia Diagnosis. *JAMA Netw Open*. 2023;6(10):e2338080. doi:10.1001/jamanetworkopen.2023.38080
 39. Sabatini S, Martyr A, Hunt A, et al. Comorbid health conditions and their impact on social isolation, loneliness, quality of life, and well-being in people with dementia: longitudinal findings from the IDEAL programme. *BMC Geriatr*. 2024;24(1):23. doi:10.1186/s12877-023-04601-x
 40. Dumitrescu L, Mahoney ER, Mukherjee S, et al; Alzheimer's Disease Neuroimaging Initiative (ADNI); Alzheimer's Disease Genetics Consortium (ADGC), A4 Study Team. Genetic variants and functional pathways associated with resilience to Alzheimer's disease. *Brain*. 2020;143(8):2561-2575. doi:10.1093/brain/awaa209
 41. Hart H, Radua J, Nakao T, Mataix-Cols D, Rubia K. Meta-analysis of functional magnetic resonance imaging studies of inhibition and attention in attention-deficit/hyperactivity disorder: exploring task-specific, stimulant medication, and age effects. *JAMA Psychiatry*. 2013;70(2):185-198. doi:10.1001/jamapsychiatry.2013.277
 42. Pievsky MA, McGrath RE. The neurocognitive profile of attention-deficit/hyperactivity disorder: a review of meta-analyses. *Arch Clin Neuropsychol*. 2018;33(2):143-157. doi:10.1093/arclin/acx055
 43. Faraone SV, Bellgrove MA, Brikell I, et al. Attention-deficit/hyperactivity disorder. *Nat Rev Dis Primers*. 2024;10(1):11. doi:10.1038/s41572-024-00495-0
 44. Faraone SV, Banaschewski T, Coghill D, et al. The World Federation of ADHD International Consensus Statement: 208 evidence-based conclusions about the disorder. *Neurosci Biobehav Rev*. 2021;128:789-818. doi:10.1016/j.neubiorev.2021.01.022
 45. Cherkasova MV, Roy A, Molina BSG, et al. Review: adult outcome as seen through controlled prospective follow-up studies of children with attention-deficit/hyperactivity disorder followed into adulthood. *J Am Acad Child Adolesc Psychiatry*. 2022;61(3):378-391. doi:10.1016/j.jaac.2021.05.019
 46. Ribasés M, Mitjans M, Hartman CA, et al. Genetic architecture of ADHD and overlap with other psychiatric disorders and cognition-related phenotypes. *Neurosci Biobehav Rev*. 2023;153:105313. doi:10.1016/j.neubiorev.2023.105313
 47. Vermunt L, Sikkes SAM, van den Hout A, et al; Alzheimer Disease Neuroimaging Initiative; AIBL Research Group; ICTUS/DSA study groups. Duration of preclinical, prodromal, and dementia stages of Alzheimer's disease in relation to age, sex, and APOE genotype. *Alzheimers Dement*. 2019;15(7):888-898. doi:10.1016/j.jalz.2019.04.001
 48. Cortese S, Faraone SV, Bernardi S, Wang S, Blanco C. Gender differences in adult attention-deficit/hyperactivity disorder: results from the National Epidemiologic Survey on Alcohol and Related Conditions (NESARC). *J Clin Psychiatry*. 2016;77(4):e421-e428. doi:10.4088/JCP.14m09630
 49. Brod M, Schmitt E, Goodwin M, Hodgkins P, Niebler G. ADHD burden of illness in older adults: a life course perspective. *Qual Life Res*. 2012;21(5):795-799. doi:10.1007/s11366-011-9981-9
 50. Michielsen M, Semeijn E, Comijs HC, et al. Prevalence of attention-deficit hyperactivity disorder in older adults in the Netherlands. *Br J Psychiatry*. 2012;201(4):298-305. doi:10.1192/bjp.bp.111.101196